A Personal Note on Rare Event Simulation

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Section 1: Introduction

Rare event simulations focus on estimating the probability of events that occur with very low probability but can have significant consequences. Because these events are so unlikely, traditional Monte Carlo simulations become inefficient, requiring an enormous number of samples to observe even a few occurrences of the rare event.

Suppose that we have an event with a probability of 10^{-6}. Using Monte Carlo simulation will need millions of samples to approximate the probability. Also, the variance of the estimate becomes very high, leading to unreliable results.

An example of this is in air traffic management. If we want to simulate the interaction between aircraft/drones, given the uncertainties in the CNS (Communication, Navigation, Surveillance) system and given the horizontal separation, if the bar for safety is very large, this might be extremely rare, so that the probability of having an intrusion is very small. Thus, we need the technique of rare event simulation.

1.1 Mathematical Foundation

Suppose A is a rare event, and you want to estimate $P(A)$. A standard Monte Carlo simulation will be:

$$\hat{P}(A) = \frac{1}{N}\sum_{i=1}^{N} 1_A(X_i) \tag{1}$$

where $1_A(X_i)$ is an indicator function that equals 1 if $X_i \in A$, 0 otherwise.

The problem is, for rare events, $1_A(X_i)$ is almost always zero, making this Monte Carlo method inefficient.

Moreover, the variance of $\hat{P}_{MC}(A)$ is high:

$$\mathrm{Var}(\hat{P}_{MC}(A)) = \frac{P(A)(1-P(A))}{N} \tag{2}$$

When $P(A)$ is very small, then:

$$P(A)(1-P(A)) \approx P(A) \tag{3}$$

Thus:

$$\mathrm{Var}(\hat{P}_{MC}(A)) \approx \frac{P(A)}{N} \tag{4}$$

If $P(A)$ is small, then a small M can already give a low variance, isn't it? But wait, let us check the coefficient of variation! **Coefficient of Variation (CV)** is a normalized measure of dispersion, or also known as **Relative Standard Deviation (RSD)**, defined as:

$$CV = \frac{\sigma}{\mu} \tag{5}$$

This metric gives a sense of how large the variability is relative to the size of the mean. A high CV indicates that the data points vary widely compared to the mean, while a low CV suggests that the data are tightly clustered around the mean.

In our Monte Carlo simulation, the mean and standard deviations are:

$$\mu = E[\hat{P}_{MC}(A)] = P(A) \tag{6}$$

$$\sigma = \sqrt{\mathrm{Var}(\hat{P}_{MC}(A))} = \sqrt{\frac{P(A)(1-P(A))}{N}} \tag{7}$$

Thus:

$$CV = \frac{\sqrt{P(A)(1-P(A))/N}}{P(A)} \tag{8}$$

For a rare event:

$$CV = \frac{\sqrt{1-P(A)}}{\sqrt{N}\cdot P(A)} \approx \frac{1}{\sqrt{N}\cdot P(A)} \tag{9}$$

Now, let's do a comparison between a non-rare event vs a rare event. Suppose that we have $P(A) = 0.5$ and $N = 10^3$, then:

$$\mathrm{Var}(\hat{P}_{MC}(A)) = \frac{0.5(1-0.5)}{10^3} = 0.25 \times 10^{-3} \tag{10}$$

$$CV = \frac{\sqrt{0.25 \times 10^{-3}}}{0.5} \approx 0.0316 = 3.16 \tag{11}$$

Thus, the standard deviation is only 3.16% of the estimated probability, making the estimator from Monte Carlo very reliable.

Another case: $P(A) = 0.9$, $N = 10^3$:

$$\mathrm{Var}(\hat{P}_{MC}(A)) = \frac{0.09}{1000} = 9 \times 10^{-5} \tag{12}$$

$$CV = \frac{\sqrt{0.025}}{0.9} \approx 0.0105 = 1.05 \tag{13}$$

Thus, the standard deviation is only 1.05% of the estimate, good!

Now, suppose that we have $P(A) = 10^{-6}$, $N = 10^3$:

$$\mathrm{Var} = \frac{10^{-6}}{10^3} = 10^{-9} \tag{14}$$

$$CV = \frac{1}{\sqrt{N}\cdot P(A)} = \frac{1}{\sqrt{10^3} \times 10^{-6}} = 0.0316 \times 10^6 \tag{15}$$

The standard deviation is 316% of the estimate! It is incredibly unreliable unless we increase the sample size.

Key takeaway: Monte Carlo simulations are efficient for non-rare events. However, when we have a rare event, one with a very small $P(A)$, the Monte Carlo solution is inefficient and unreliable for a small sample size. We need an extremely large N for it to work. To handle this issue, we need specialized techniques in Rare Event Simulation, such as **Importance Sampling** and **Splitting Methods**.

Section 2: Importance Sampling

Importance sampling is a powerful technique used to improve the efficiency of Monte Carlo simulations, especially when estimating probabilities of rare events. The core idea is to change the probability distribution from which we sample, making the rare event more frequent in the samples. This leads to more informative simulations without needing an excessive number of samples.

Let's recall the variance and coefficient of variation (CV) in Monte Carlo simulation:

$$\hat{P}_{MC}(A) = \frac{1}{M}\sum_{i=1}^{M} 1_A(X_i), \quad X_i \sim f(x) \tag{16}$$

$$\mathrm{Var}(\hat{P}_{MC}(A)) = \frac{P(A)(1-P(A))}{M} \tag{17}$$

$$CV = \frac{\sqrt{1-P(A)}}{\sqrt{M}P(A)} \tag{18}$$

2.1 Variance in Importance Sampling

Variance in importance sampling is different! Let's start with the idea: Instead of sampling from $f(x)$, we sample from a different distribution $g(x)$, which makes the rare event more frequent. To correct for the change in the sampled data, we multiply by a weight, known as the likelihood ratio or importance weight.

$$P(A) = \int 1_A(x) f(x)\, dx = \int 1_A(x) \frac{f(x)}{g(x)} g(x)\, dx \tag{19}$$

Thus, the importance sampling estimator is:

$$\hat{P}_{IS}(A) = \frac{1}{N}\sum_{i=1}^{N} 1_A(X_i) \frac{f(X_i)}{g(X_i)}, \quad X_i \sim g(x) \tag{20}$$

where:

- $f(x)$ is the original probability density function (PDF).
- $g(x)$ is the importance sampling PDF.
- $\frac{f(x)}{g(x)}$ is the importance weight, correcting for the change in the sampling distribution.

Note: The mean of the estimator remains:

$$E_g[\hat{P}_{IS}(A)] = P(A) \tag{21}$$

Thus, the variance is:

$$\mathrm{Var}(\hat{P}_{IS}(A)) = \frac{1}{N}\mathrm{Var}_g\left[1_A(x)\frac{f(x)}{g(x)}\right] \tag{22}$$

Since the samples X_i are independent and identically distributed (i.i.d.), the variance of the sum scales with $\frac{1}{N}$:

$$\mathrm{Var}(\hat{P}_{IS}(A)) = \frac{1}{N}\mathrm{Var}_g\left[1_A(x)\frac{f(x)}{g(x)}\right] \tag{23}$$

We can write it as:

$$\mathrm{Var}\left(\frac{1}{N}\sum_{i=1}^{N} Y_i\right) = \frac{1}{N^2}\sum_{i=1}^{N}\mathrm{Var}(Y_i) = \frac{\mathrm{Var}(Y)}{N} \tag{24}$$

where:

$$Y_i = 1_A(x)\frac{f(x)}{g(x)} \tag{25}$$

Thus, the variance of a random variable Y is:

$$\mathrm{Var}(Y) = E_g[Y^2] - (E_g[Y])^2 \tag{26}$$

$$\mathrm{Var}(\hat{P}_{IS}(A)) = \frac{1}{N}\left(E_g\left[1_A(x)\frac{f(x)}{g(x)}\right]^2 - P(A)^2\right) \tag{27}$$

2.2 Finding the Optimal Importance Sampling Distribution

To obtain the optimal $g(x) \to g^*(x)$, we need to make the variance as small as possible. So, we want to minimize:

$$\int_A \frac{f(x)^2}{g(x)}\,dx - P(A)^2 \tag{28}$$

Using Lagrange multipliers, we introduce the constraint:

$$\int_A g(x)\,dx = 1 \tag{29}$$

$$\int_A \frac{f(x)^2}{g(x)}\,dx + \lambda\left(\int_A g(x)\,dx - 1\right) \tag{30}$$

Setting the first derivative to zero:

$$\frac{f(x)^2}{g(x)^2} = \lambda \tag{31}$$

Thus,

$$g(x)^2 = \frac{f(x)^2}{\lambda} \Rightarrow g(x) = \frac{f(x)}{\sqrt{\lambda}} \tag{32}$$

Using the constraint:

$$\int_A g(x)\,dx = 1 \tag{33}$$

$$\int_A \frac{f(x)}{\sqrt{\lambda}}\,dx = 1 \tag{34}$$

$$\frac{1}{\sqrt{\lambda}} \int_A f(x)\, dx = 1 \tag{35}$$

$$\frac{1}{\sqrt{\lambda}} P(A) = 1 \tag{36}$$

$$\lambda = P(A)^2 \tag{37}$$

Thus, the optimal importance sampling distribution is:

$$g^*(x) = \frac{f(x)}{P(A)} \tag{38}$$

Since $g^*(x)$ must be zero outside of A, we introduce the indicator function:

$$g^*(x) = \frac{1_A(x) f(x)}{P(A)} \tag{39}$$

However, $P(A)$ is not known, so it is impractical. But if we choose the correct $g(x)$, we can significantly reduce the variance.

2.3 Example

Suppose that we want to estimate $P(X > 5)$ from a normal distribution $\mathcal{N}(0, 1)$ using importance sampling from $\mathcal{N}(5, 1)$. In a standard Monte Carlo simulation:

$$P(X > 5) = \int_5^\infty f(x)\, dx \tag{40}$$

where $f(x)$ is the probability density function of a standard normal distribution:

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \tag{41}$$

To improve the estimation, we shift the mean to 5 for importance sampling:

$$g(x) = \frac{1}{\sqrt{2\pi}} e^{-(x-5)^2/2} \tag{42}$$

The importance weight is:

$$w(x) = \frac{f(x)}{g(x)} = e^{\frac{-x^2+(x-5)^2}{2}} \tag{43}$$

$$= e^{-5x+25/2} \tag{44}$$

Thus, the importance sampling estimator becomes:

$$\hat{P}_{IS}(X > 5) = \frac{1}{M} \sum_{i=1}^{M} 1_{X_i > 5} e^{-5X_i + 25/2}, \quad X_i \sim \mathcal{N}(5, 1) \tag{45}$$

The expected value:

$$E_g[\hat{P}_{IS}(X > 5)] = \int_5^\infty e^{-5x+25/2} \frac{1}{\sqrt{2\pi}} e^{-x^2/2}\, dx \tag{46}$$

$$= \int_5^\infty \frac{1}{\sqrt{2\pi}} e^{-x^2/2}\, dx \tag{47}$$

This confirms that importance sampling in this case is unbiased.

Section 3: Splitting Method

Splitting Methods (also called *Multilevel Splitting* or *Subset Simulation*) are a class of variance reduction techniques designed to estimate the probability of rare events efficiently. Unlike importance sampling, which changes the sampling distribution, splitting methods decompose the rare event into a sequence of more frequent intermediate events, allowing for a progressive refinement of the estimation.

The key idea of splitting methods is to divide the event into multiple regions. Instead of using a large number of independent trajectories, we can adaptively increase the number of samples reaching the rare event region.

The main steps of splitting methods:

- **Break down** the rare event into intermediate levels: $A_1 \subset A_2 \subset ... \subset A_m = A$.
- **Simulate trajectories**, filtering out those that reach the next level.
- **Replicate (split)** promising trajectories, effectively increasing the probability of reaching the final rare event.

3.1 Mathematical Formulation

Suppose we want to estimate the probability of a rare event A:

$$P(A) = P(X \in A) \tag{48}$$

where X follows some distribution $f(x)$. The challenge is that direct Monte Carlo estimation is inefficient when $P(A)$ is very small.

To solve this, we decompose $P(A)$ into intermediate levels:

$$A_1 \subset A_2 \subset A_3 \subset ... \subset A_m = A \tag{49}$$

Using conditional probability:

$$P(A) = P(A_m) = P(A_1)P(A_2|A_1)P(A_3|A_2)...P(A_m|A_{m-1}) \tag{50}$$

which can be rewritten as:

$$P(A) = \prod_{i=1}^{m} P(A_i|A_{i-1}) \tag{51}$$

Where:

- $P(A_1)$ is the probability of reaching the first level.
- $P(A_i|A_{i-1})$ is the probability of reaching level A_i given that level A_{i-1} was reached.

Thus, instead of directly estimating $P(A)$, we estimate each intermediate probability $P(A_i|A_{i-1})$, which is much larger than $P(A)$.

For a Monte Carlo Simulation, we estimate each probability $P(A_i|A_{i-1})$ using:

$$\hat{P}(A) = \prod_{i=1}^{m} \hat{P}(A_i|A_{i-1}) \tag{52}$$

where:

$$\hat{P}(A_i|A_{i-1}) = \frac{\text{number of samples reaching } A_i}{\text{number of samples reaching } A_{i-1}} \tag{53}$$

3.2 Variance Reduction in Splitting Methods

One important question: How does this technique reduce variance?

For standard Monte Carlo estimation, the variance is:

$$\mathrm{Var}(\hat{P}_{MC}(A)) = \frac{P(A)(1-P(A))}{N} \tag{54}$$

The coefficient of variation (CV) is:

$$CV = \frac{\sqrt{\mathrm{Var}(\hat{P}_{MC}(A))}}{E[\hat{P}_{MC}(A)]} = \frac{\sqrt{1-P(A)}}{\sqrt{N}P(A)} \tag{55}$$

For rare events ($P(A) \ll 1$):

$$CV \approx \frac{1}{\sqrt{N}P(A)} \tag{56}$$

For splitting methods, the estimate becomes:

$$\hat{\theta} = \hat{P}_{MC}(A) \prod_{i=1}^{m} \hat{P}_i \tag{57}$$

Using delta-method analysis, let $\bar{X}_n = (X_1, X_2, ..., X_n)$, a set of estimators converging to a vector of parameters $\bar{\pi} = (\pi_1, ..., \pi_m)$. Then:

$$\sqrt{n}(\bar{X}_n - \bar{\pi}) \xrightarrow{d} \mathcal{N}(0, \Sigma) \tag{58}$$

where Σ is the covariance matrix. Applying a smooth function $g(\cdot)$:

$$\sqrt{n}\left(g(\bar{X}_n) - g(\bar{\pi})\right) \xrightarrow{d} \mathcal{N}(0, \nabla g(\bar{\pi})^T \Sigma \nabla g(\bar{\pi})) \tag{59}$$

For splitting methods:

$$g(p_1, \ldots, p_m) = \prod_{i=1}^{m} p_i \tag{60}$$

Thus, if $\bar{\pi} = (p_1, ..., p_m)$ is the true vector and $\hat{\pi} = (\hat{p}_1, ..., \hat{p}_m)$ is the estimator, then:

$$\hat{\theta} = g(\hat{p}) \tag{61}$$

The gradient:

$$\nabla g(\bar{p}) = \left(\frac{\partial}{\partial p_1} \prod_{i=1}^{m} p_i, \ldots, \frac{\partial}{\partial p_m} \prod_{i=1}^{m} p_i \right) \tag{62}$$

For each component:

$$\frac{\partial}{\partial p_i} \prod_{i=1}^{m} p_i = \prod_{j \neq i} p_j \tag{63}$$

Thus:

$$\nabla g(\bar{p}) = \theta \left(\frac{1}{p_1}, ..., \frac{1}{p_m} \right) \tag{64}$$

The variance is then:

$$\mathrm{Var}(\hat{\theta}) \approx \nabla g(\bar{p})^T \Sigma \nabla g(\bar{p}) = \theta^2 \sum_{i=1}^{m} \frac{\mathrm{Var}(\hat{p}_i)}{p_i^2} \tag{65}$$

assuming that $\mathrm{Cov}(\hat{p}_i, \hat{p}_j) \approx 0$ for $i \neq j$.

The variance reduction comes from the fact that each intermediate probability $P(A_i|A_{i-1})$ is much larger than the total probability $P(A)$. Since variance is inversely proportional to probability in Monte Carlo estimation, breaking the probability into conditional steps significantly increases the precision of each term, leading to a lower overall variance.

Additionally, the splitting method reallocates samples efficiently, ensuring that most of the computational effort is focused on the more probable intermediate events rather than wasting samples on rare failures. This focused sampling dramatically reduces the spread of the estimate.

Applying the variance result to the coefficient of variation:

$$CV_{\text{splitting}} = \frac{\sqrt{\mathrm{Var}(\hat{\theta})}}{E[\hat{\theta}]} \tag{66}$$

Using our previous result:

$$CV_{\text{splitting}} = \frac{\theta\sqrt{\sum_{i=1}^{m} \frac{\mathrm{Var}(\hat{p}_i)}{p_i^2}}}{\theta} = \sqrt{\sum_{i=1}^{m} \frac{\mathrm{Var}(\hat{p}_i)}{p_i^2}} \tag{67}$$

Since each p_i is significantly larger than $P(A)$, the variance term inside the sum is much smaller compared to the standard Monte Carlo case, leading to a smaller overall CV. This shows that **splitting methods not only reduce variance but also improve the stability of the estimator**.

3.3 Example: Brownian Motion with a Reflecting Barrier

Suppose we have a Brownian motion with drift and want to estimate the probability of reaching the upper boundary b before hitting a lower boundary a. The process is described by:

$$X_t = B_t + \mu t \tag{68}$$

where:

- B_t is standard Brownian motion.
- $\mu > 0$ is a negative drift (biasing the process toward a).

If $H = \inf\{t \geq 0 : X_t \notin [a, b]\}$, then the probability of reaching b before a is:

$$P(X_H = b) = e^{-\mu(b-a)} \frac{\sinh(\mu(X-a))}{\sinh(\mu(b-a))} \tag{69}$$

To simulate this process computationally, we discretize the equation:

$$B_{t+\Delta t} = B_t + \sqrt{\Delta t} Z \tag{70}$$

where $Z \sim \mathcal{N}(0, 1)$. If we introduce drift, the equation becomes:

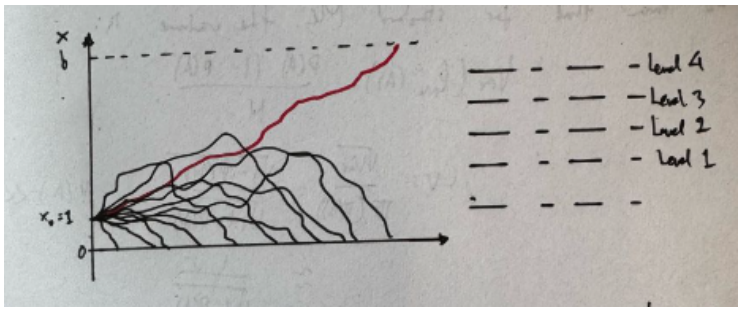
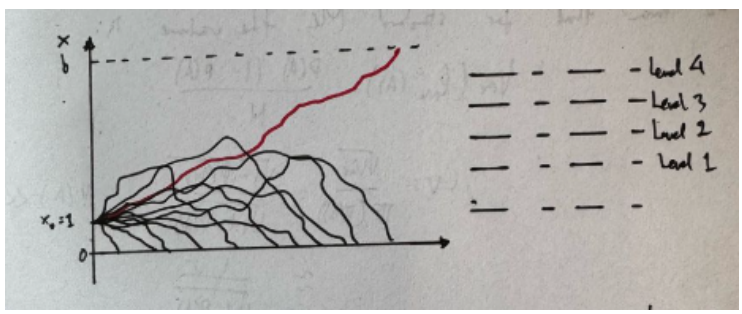


Figure 1: Multilevel Splitting Illustration

$$B_{t+\Delta t} = B_t + \mu\Delta t + \sqrt{\Delta t}Z \tag{71}$$

Thus, we can use this discretized version to implement the splitting method in a simulation.

The simulation will look like this:



In a classical splitting method, we divide the process into different levels. Each level represents a checkpoint that helps break down the rare event probability into more frequent conditional probabilities. Instead of estimating $P(X_H = b)$ directly, we express it as:

$$P(X_H = b) = P(X_1|X_0)P(X_2|X_1)P(X_3|X_2)\dots P(X_H|X_{H-1}) \tag{72}$$

Each term represents the conditional probability of reaching the next level before returning to a. These intermediate probabilities are much larger than directly estimating $P(X_H = b)$.

Then, we count the frequency of each level and calculate how many of them reach the next level:

$$\hat{P}(X_{i+1}|X_i) = \frac{\text{number of successful trajectories}}{\text{number of samples at } X_i} \tag{73}$$

The choice of how many "duplicates" per successful trajectory is up to the user.

Finally:

$$\hat{P}(X_H|X_0) = \prod_{i=0}^{m-1} \hat{P}(X_{i+1}|X_i) \tag{74}$$

References